Practical HW 1

Machine Learning 2

April 2026

Exploring youth substance use

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# Research Question:

How well can youth demographics, family life, and experiences predict alcohol usage in a month period,
whether the youth has ever used marijuana, and cigarette usage frequency?
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Utilizing: Decision Trees

Specifically...

Binary classification

Multi-class classification

Regression classification

Data description

From the NSDUH dataset, the youth data was selected (responses from ages under 18). Variables included substance use frequency, youth experiences (e.g. perceived peer disapproval, grade average, frequency of parents checking their homework), and demographics (e.g. income level, race, sex).

Target Outcomes:

1. Binary Classification: MRJFLAG (ever used marijuana)
2. Multi-Class Classification: ALCMDAYS (# of days used alcohol in past month)
3. Regression Classification: IRCIGFM (cigarette frequency in past month)

Predictors:

- IFATHER (categorical): whether the youth's father is in the household or not
- POVERTY3 (categorical): whether the youth is living in poverty, up to two times the federal poverty level, or more than two times the federal poverty level.
- IRSEX (binary): whether the youth is male or female
- NEWRACE2 (categorical): the youth's race/ethnicity (7 possible categories)
- PRCHORE2 (binary): how often the youth's parents make them do chores around the house in the past year
- PRLMTTV2 (binary): how often the youth's parents limit their amount of TV in the past year
- PARCHKHW (binary): how often the youth's parents check if their homework is completed in the past year
- YFLTMRJ2 (binary): how the youth feels about their peers trying marijuana
- RLGATTD (binary): number of times the youth has attended religious services in the past year
- AVGGRADE (binary): the youth's grade average for the last grading period completed

Modeling Approach

For each classification type:

1. Baseline decision trees
    a. Evaluation metrics
2. Tuning/Pruning with cross-validation
    a. Evaluation metrics
3. Comparisons of baseline and tuned trees

Then,

4. Ensemble methods (Random forests)

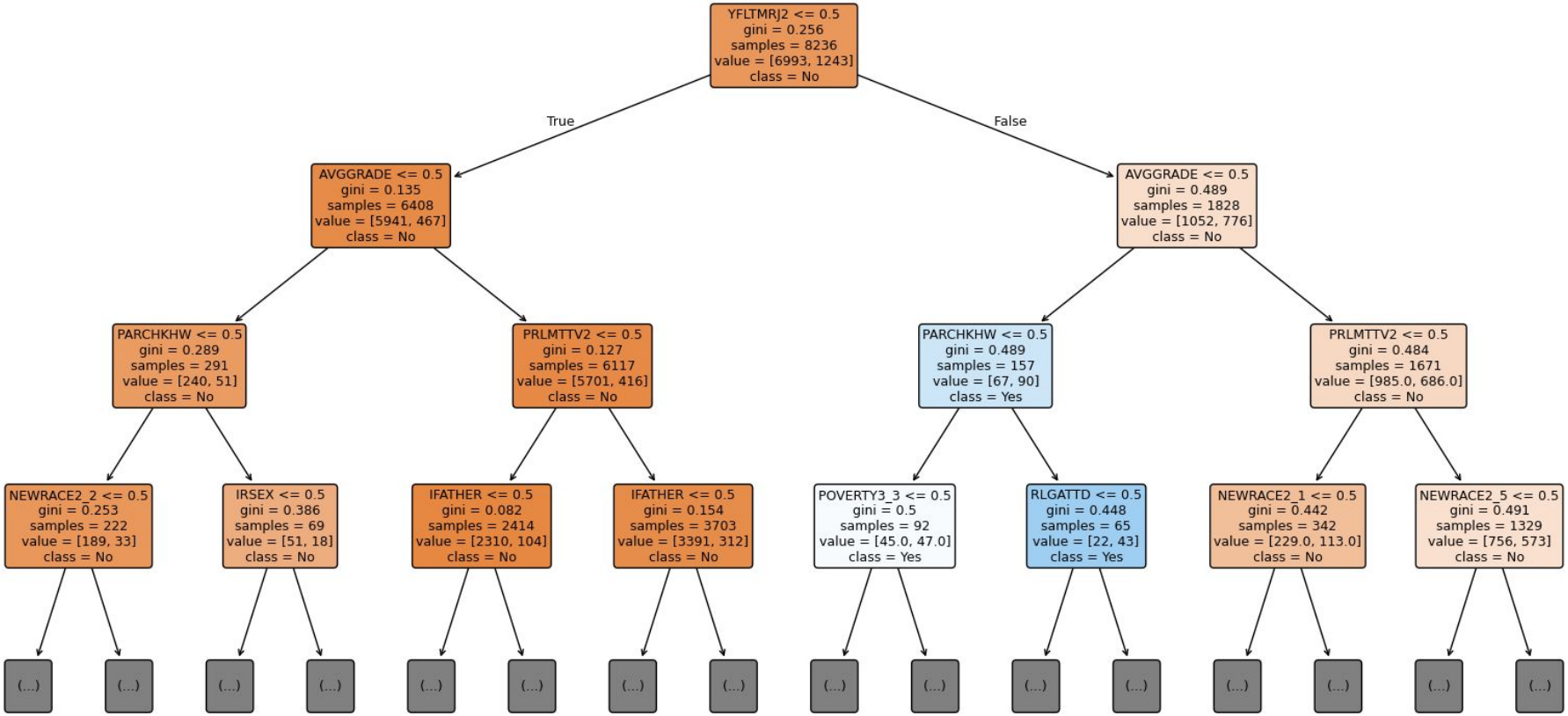
Metrics evaluated:

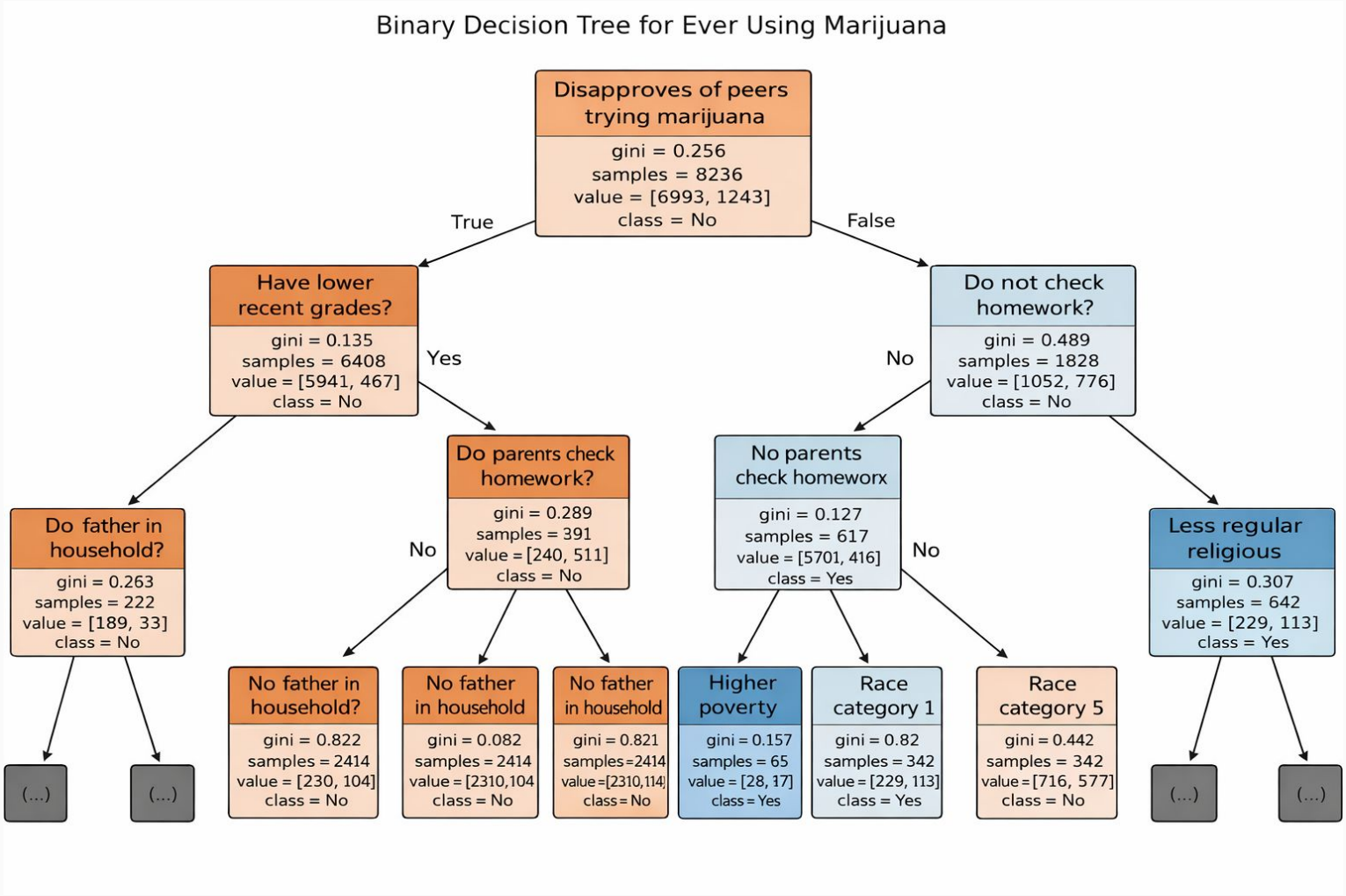
Binary classification: accuracy, precision, f1 score, recall

Multi-class classification: accuracy, classification report

Regression classification: mean squared error, residual sum of squares, r-squared

Binary Decision Tree for Ever Using Marijuana




Binary Decision Tree for Ever Using Marijuana
Disapproves of peers trying marijuana
gini = 0.256
samples = 8236
value = [6993, 1243]
class = No
True
False
Have lower recent grades?
gini = 0.135
samples = 6408
value = [5941, 467]
class = No
Yes
Do not check homework?
gini = 0.489
samples = 1828
value = [1052, 776]
class = No
No
Do father in household?
gini = 0.263
samples = 222
value = [189, 33]
class = No
Do parents check homework?
gini = 0.289
samples = 391
value = [240, 511]
class = No
No
No parents check homeworx
gini = 0.127
samples = 617
value = [5701, 416]
class = Yes
No
Less regular religious
gini = 0.307
samples = 642
value = [229, 113]
class = Yes
(...)
(...)
No father in household?
gini = 0.822
samples = 2414
value = [230, 104]
class = No
No father in household
gini = 0.082
samples = 2414
value = [2310,104
class = No
No father in household
gini = 0.821
samples=2414
value=[2310,114]
class=No
Higher poverty
gini = 0.157
samples = 65
value = [28, 17]
class = Yes
Race category 1
gini = 0.82
samples = 342
value = [229, 113]
class = Yes
Race category 5
gini = 0.442
samples = 342
value = [716, 577]
class = No
(...)
(...)

The plot above shows the binary decision tree used to predict whether a youth ever used marijuana.

The first split is on YFLTMRJ2 <= 0.5, which measures how the youth feels about their peers trying marijuana. This path follows the left branch, which means ***. The second split is on AVGGRADE <= 0.5, which measures the youth's **. This path follows the left branch, which means *. The third split is on PARCHKHW <= 0.5, which measures how often the youth's parents check if they completed their homework.

Discussion

Class imbalance is a major concern that needs to be accounted for. Models oftentimes, including in this investigation, struggle to accurately classify minority classes. The models often just default classifying to the dominant class. The accuracy percent still oftentimes, including in this investigation, is high enough to be interpreted as adequately accurate classification is occurring. A deeper look into the accuracy by class must be done to better accurately measure classification and identify class imbalance.

Conclusion

Analyzing demographics and experiences youths have had can possibly give us some insight into their substance usage.

The multi-class and binary classification models both have a common top level predictor: how the youth feels about their peers using marijuana.

Limitations are present with this investigation, considering some variables are redundant (predictor and target are too closely related or two predictors capture the same idea) and the code was not rigorously explored.